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Anime Score Prediction

Introduction

Our project was centered around the following research question: Can various aspects of Japanese animated media (animes), such as duration and release year, accurately predict the show's rating out of 10? The motivation behind our project is to create a machine learning model that allows users to predict the rating of an anime to better guide their viewing choices. This project utilizes a dataset specifically tailored for anime, which is a popular genre of cartoons, primarily produced in Japan. Since this type of media is so popular across the world, there is significant impact to be made from offering a better recommendation tool for those who watch anime.

Methods

The dataset our group used originated from Kaggle and had information on thousands of animes. We aimed to predict the score (out of 10) for each anime, which was represented in the data set by the score column. As this was the variable we aimed to predict, any rows with missing score values were dropped. The dataset originally had 35 feature columns, many of which had to be altered or dropped. Certain columns, such as the duration and air date, had to be cleaned to obtain these values in a numerical format. Other columns, such as genres and type, had to be one-hot encoded to retain their information. Finally, many columns had to be dropped due to formatting or relevance. Columns such as the shows' ID, English name, and Japanese name were dropped because they were either not transformable or useful for predictions. Some columns were dropped because they would have been too much information to one-hot encode, which includes the producers, licensors, and studios columns. Also, these columns had many missing values. Finally, there were columns that gave the number of people who scored the anime a certain way. An example of this is the "Score-10" column, which gave the number of people who scored that anime a 10/10. These columns were dropped because the shows actual score is likely an average of these columns, so using them might skew the predictions the model makes.

We used four models in total: (1) Ridge Regression, (2) Support Vector Machine, (3) Random Forest, (4) and a Neural Network. To train and evaluate each model, we further split the training dataset into smaller train and test sets. Before using this data, certain models needed specific preprocessing steps. To run the Neural Network, we had to scale

the data using a standard scaler, then convert our data into tensors. The Random Forest model required no preprocessing. The use of a standard scaler on the data was also implemented for the Support Vector Regression (SVR) method and the Ridge Regression model. To optimize each model's hyperparameters, we used Grid Search CV or K Fold Validation. The metric we used to compare the models was Mean Square Error because this was a regression task. Additionally, each model had certain unique scoring metrics.

Results

Model	MSE
Neural Network	0.234
SVM	0.229
Random Forest	0.237
Ridge Regression	0.288

Based on the table showcasing the results from the four models used on the training data, Mean Squared Error (MSE) was used to compare the overall performance for each model. SVR achieved the lowest error with an MSE of **0.229**, indicating that it had the strongest predictive performance. The Neural Network also performed well with an MSE of **0.234**, demonstrating that it was able to capture nonlinear patterns effectively. The Random Forest produced a similarly competitive MSE of **0.237**, confirming the strength of ensemble tree methods for this task. Ridge Regression, while the simplest model, yielded the highest error, an MSE of **0.288**, reflecting its limited capacity to assess/model complex relationships. Overall, this demonstrated that all four models had different predictive capabilities for generating scores based on the feature columns. The model which performed the best was the SVR, since it had the lowest MSE. After the SVR method was trained on the training data and evaluated on the test data, a MSE of 0.217 was found.

Discussion

Overall, the SVR model performed the best as it had the lowest MSE. This means that if a user wanted to accurately predict the score of an anime, SVR would be the ideal option. SVR might have outperformed the other models because it can handle high-dimensional data and not succumb to the "Curse of Dimensionality". SVR also might have outperformed the other models because it was well suited to handle the many binary columns that this dataset included after one-hot encoding. One limitation of this project is that many rows had to be dropped because the score value was missing. Another limitation about the dataset was that it was heavily skewed towards more recent years.

References:

[1] H. Valdivieso, "Anime Recommendation Database 2020," Kaggle, 2020. [Online]. Available: <https://www.kaggle.com/datasets/hernan4444/anime-recommendation-database-2020> [Accessed Nov. 12, 2025].